

SIMULATION BUILDING IN PHYSICAL SCIENCE

Solid-state physics largely employs simulation techniques for modelling. By taking data from experiments, crystal lattice structures can be constructed easily.

Figure 7 shows one such three-dimensional array of lattice points. The properties of the crystal structure can be inferred by building them. Simulations will give us a clear picture of the crystal properties by considering its primitives.

So is the case with complex bodies. It is found that Jupiter is accompanied, in its orbit, by two groups of asteroids that precede it and follow it at an angular distance of $\pi/3$ (see Figure 8). By building simulations we can show that these are positions of stable equilibrium. A Runge-Kutta procedure with automatic step control can be used for analysing the data from Table 2.

Data for analysing a Runge-Kutta procedure with automatic step control

	Sun	Jupiter	Trojan 1	Trojan 2
Mass	1	0.001	0	0
x	0	0	-4.50333	4.50333
y	0	5.2	2.6	2.6
z	0	0	0	0
V_x	0	-2.75674	-1.37837	-1.37837
V_y	0	0	-2.38741	2.38741
V_z	0	0	0	0

Table 2

Using the computational procedure we can get a prediction as shown in Figure 9.

You can also try solving simple problems of the following form:

$$\frac{d^2x}{dy^2} = F\left(x, y, \frac{dy}{dx}\right)$$

The Runge-Kutta procedure is quite sufficient to handle these types of problems, provided you have enough data.

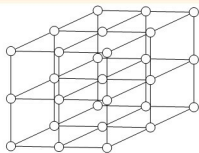


Figure 7: A 3D array of lattice points

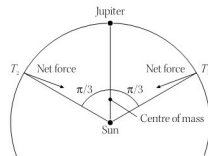


Figure 8: Jupiter is accompanied by two groups of asteroids that precede and follow it at an angular distance of $\pi/3$

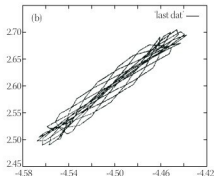


Figure 9: This prediction can be made using computational procedures

```
make arrays L[1 ...s1 + 1] and R[1 ...s2 + 1]
```

```
for i ← 1 to s1
```

```
do L[i] ← A[p + i - 1]
```

```
for j ← 1 to s2
```

```
do R[j] ← A[q + j]
```

```
L[s1 + 1] ← ∞
```

```
R[s2 + 1] ← ∞
```

```
i ← 1
```

```
j ← 1
```

```
for k ← p to r
```

```
do if L[i] < R[j]
```

```
then A[k] ← L[i]
```

```
i ← i + 1
```

```
else A[k] ← R[j]
```

```
j ← j + 1
```

Let's look at an example to comprehend the idea better. I found the following problem in a book that deals with problem solving. A part of the problem involves arranging an array of values [5, 2, 4, 7, 1, 3, 2, 6]. This will be our initial (given) array. Now, we can use our methodology to solve the problem. Figure 3 shows the consecutive division and merging processes.

We can imagine the process in many different ways. If the set is random, then we could have another array with the same elements in a different order. The solution is shown in Figures 4, 5 and 6. Here, Figures 4 and 5 represent the initial steps and Figure 6 represents the final output. It should be noted that the intermediate stages are not shown here, as they are similar steps that can be envisaged in one's mind. The step-by-step procedure will also throw ample light on the subs L and R that are created.

Once you are done with simple algorithms, you can apply the same 'black box representation' idea (that you employ while writing